# Module 6a slides: Multiple Oral Doses (worked examples)

Module 6 - Multiple Dosings · Lecture slides, worked examples · 28 Sep 2026

Name _____ Date _____

**Example 1** An adult male (75 kg) received 250 mg tetracycline hydrochloride orally every 8 hours for 2 weeks. Tetracycline hydrochloride is about 75% bioavailable and has a $V_D$ of 1.5 L/kg. The t½ is about 10 hours and the $k_a$ is 0.9 $hr^{-1}$. When does the peak of the first dose occur?

**Example 1** For the tetracycline regimen (250 mg orally, F 0.75, $V_D$ 112.5 L, k 0.0693 $hr^{-1}$, $k_a$ 0.9 $hr^{-1}$), the first dose peaks at 3.09 hours. What is $C_{max}$ of the first dose?

**Example 1** For the tetracycline regimen (250 mg orally every 8 hours, k 0.0693 $hr^{-1}$, $k_a$ 0.9 $hr^{-1}$), when does the peak occur at steady state?

**Example 1** For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, $V_D$ 112.5 L, k 0.0693 $hr^{-1}$), the steady-state peak occurs at 2.06 hours. What is $C_{max}$ at steady state?

**Example 1** For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, $V_D$ 112.5 L, k 0.0693 $hr^{-1}$, $k_a$ 0.9 $hr^{-1}$), what is $C_{min}$ at steady state?

**Example 1** For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, $V_D$ 112.5 L, k 0.0693 $hr^{-1}$), what is Cavg at steady state?